Algebraic Ideal Vapor-Compression Cycle for R32: Governing Equations (No Simultaneous Solver)

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1 Why an Equation of State Is Still Required

Removing the simultaneous nonlinear solver (IPOPT, as used by IDAES) does not remove the need for an equation of state (EoS). The solver and the EoS serve different roles. IDAES uses IPOPT to solve the full flowsheet — every unit model's mass and energy balances, phase-equilibrium constraints, and bounds — as one large coupled nonlinear system. That simultaneous solve is what introduces path-dependent convergence behavior, a tolerance floor (tol = 10^{-4}), and warmstart-sensitive failures unrelated to the underlying physics.

The equation of state, by contrast, is the physical model itself: it is what allows real (non-ideal) enthalpy and entropy to be computed near the saturation dome, where the refrigerant spends the entire cycle. An ideal-gas heat capacity alone cannot describe this region. Evaluating the Peng–Robinson EoS at a known temperature and pressure requires no iterative solver: the cubic in compressibility factor Z is solved directly via its closed-form/eigenvalue root (deterministic, no initial guess, no failure mode), and the departure functions built on top of it are closed-form integrals. The only root-finding anywhere in this formulation is a single well-posed one-dimensional bisection (Sec. **??**), fundamentally different in kind from a large coupled simultaneous solve.

2 Peng–Robinson Equation of State

$$P = \frac{RT}{V-b} - \frac{a(T)}{V(V+b)+b(V-b)} \tag{1}$$

with temperature-dependent attraction parameter

$$a(T) = 0.45724\,\frac{R^2T_c^2}{P_c}\,\alpha(T), \qquad \alpha(T) = \left[1+\kappa\left(1-\sqrt{T/T_c}\right)\right]^2 \tag{2}$$

$$\kappa = 0.37464 + 1.54226\,\omega - 0.26992\,\omega^2 \tag{3}$$

and size parameter

$$b = 0.07780\,\frac{R\,T_c}{P_c} \tag{4}$$

where T_c, P_c are the critical temperature and pressure and ω is the acentric factor. In compressibility-factor form, with $A = a(T)P/(RT)^2$ and $B = bP/(RT)$:

$$Z^3 - (1-B)Z^2 + \left(A - 2B - 3B^2\right)Z - \left(AB - B^2 - B^3\right) = 0 \tag{5}$$

This cubic is solved directly for its real roots; the largest admissible root ($Z > B$) is the vapor phase, the smallest is the liquid phase.

3 Departure Functions (Poling, Table 6-3; PR: $u = 2$, $w = -1$)

Real-fluid enthalpy and entropy are expressed as an ideal-gas baseline plus a departure term computed from the EoS:

$$h(T,P) = h^{\mathrm{ig}}(T) + h^{\mathrm{dep}}(T,P,Z), \qquad s(T,P) = s^{\mathrm{ig}}(T,P) + s^{\mathrm{dep}}(T,P,Z) \tag{6}$$

$$h^{\mathrm{dep}} = RT(Z-1) + \frac{T\dfrac{da}{dT} - a}{2\sqrt{2}\,b} \ln\left(\frac{Z + (1+\sqrt{2})B}{Z + (1-\sqrt{2})B}\right) \tag{7}$$

$$s^{\mathrm{dep}} = R\ln(Z-B) + \frac{\dfrac{da}{dT}}{2\sqrt{2}\,b} \ln\left(\frac{Z + (1+\sqrt{2})B}{Z + (1-\sqrt{2})B}\right) \tag{8}$$

4 Ideal-Gas Enthalpy and Entropy (Shomate)

With $t = T/1000$ and Shomate coefficients A–E:

$$h^{\mathrm{ig}}(T) = 1000\left(At + \frac{Bt^2}{2} + \frac{Ct^3}{3} + \frac{Dt^4}{4} - \frac{E}{t}\right) \tag{9}$$

$$s^{\mathrm{ig}}(T,P) = A\ln t + Bt + \frac{Ct^2}{2} + \frac{Dt^3}{3} - \frac{E}{2t^2} - R\ln\left(\frac{P}{P_{\mathrm{ref}}}\right) \tag{10}$$

5 Saturation Pressure (Ambrose–Walton, Poling §7-4)

With reduced temperature $T_r = T/T_c$ and $\tau = 1 - T_r$:

$$f^{(0)} = \frac{-5.97616\,\tau + 1.29874\,\tau^{1.5} - 0.60394\,\tau^{2.5} - 1.06841\,\tau^{5}}{T_r} \tag{11}$$

$$f^{(1)} = \frac{-5.03365\,\tau + 1.11505\,\tau^{1.5} - 5.41217\,\tau^{2.5} - 7.46628\,\tau^{5}}{T_r} \tag{12}$$

$$f^{(2)} = \frac{-0.64771\,\tau + 2.41539\,\tau^{1.5} - 4.26979\,\tau^{2.5} + 3.25259\,\tau^{5}}{T_r} \tag{13}$$

$$P_{\mathrm{sat}}(T) = P_c \exp\Big(f^{(0)} + \omega f^{(1)} + \omega^2 f^{(2)}\Big) \tag{14}$$

This is evaluated directly and requires no iteration; it is what fixes P_{evap} and P_{cond} once the saturation temperatures are specified.

6 The Cycle: Sequential, Not Simultaneous

Operating specification: evaporator saturation temperature T_{evap} fixed; condenser saturation temperature $T_{\text{cond}} = T_{\text{amb}} + \Delta T_{\text{approach}}$; superheat = subcool = 0; compressor isentropic efficiency η. Because the saturation temperatures are fixed and there is no superheat or subcooling, each state can be evaluated from information already known before it — the system is sequential rather than a coupled simultaneous system.

State 1 — Evaporator Outlet (Saturated Vapor)

$$P_{\text{evap}} = P_{\text{sat}}(T_{\text{evap}}), \qquad h_1 = h_{\text{vap}}(T_{\text{evap}}, P_{\text{evap}}), \qquad s_1 = s_{\text{vap}}(T_{\text{evap}}, P_{\text{evap}}) \tag{15}$$

Fully determined; no iteration.

State 2 — Compressor Outlet

The isentropic discharge temperature T_{2s} is the unique root of

$$s_{\text{vap}}(T_{2s}, P_{\text{cond}}) - s_1 = 0 \tag{16}$$

Since $s_{\text{vap}}(T, P_{\text{cond}})$ is monotonically increasing in T, this is a well-posed one-dimensional bisection (bracketed root find, e.g. Brent's method) — the only implicit step in the entire cycle, and categorically different from a coupled multi-equation solve: one variable, one guaranteed-monotonic function, no path dependence.

$$h_{2s} = h_{\text{vap}}(T_{2s}, P_{\text{cond}}), \qquad h_2 = h_1 + \frac{h_{2s} - h_1}{\eta} \tag{17}$$

State 3 — Condenser Outlet (Saturated Liquid)

$$P_{\text{cond}} = P_{\text{sat}}(T_{\text{cond}}), \qquad h_3 = h_{\text{liq}}(T_{\text{cond}}, P_{\text{cond}}), \qquad s_3 = s_{\text{liq}}(T_{\text{cond}}, P_{\text{cond}}) \tag{18}$$

Fully determined; no iteration.

State 4 — Valve Outlet (Isenthalpic Expansion)

$$h_4 = h_3 \tag{19}$$

a direct assignment, not a calculation. The resulting state is two-phase at P_{evap}; its vapor quality follows by linear interpolation between the saturated-liquid and saturated-vapor enthalpies at T_{evap} (already available from State 1 and the corresponding liquid evaluation), again with no root-finding required:

$$x_4 = \frac{h_4 - h_{f,4}}{h_{g,4} - h_{f,4}}, \qquad s_4 = s_{f,4} + x_4 \left(s_{g,4} - s_{f,4}\right) \tag{20}$$

7 Coefficient of Performance

$$q_{\text{evap}} = h_1 - h_4, \qquad w_{\text{comp}} = h_2 - h_1 \tag{21}$$

$$\boxed{\text{COP} = \frac{q_{\text{evap}}}{w_{\text{comp}}} = \frac{h_1 - h_4}{h_2 - h_1}} \tag{22}$$

Because COP is a ratio of enthalpy *differences*, any constant reference-state offset in h^{ig} or s^{ig} cancels exactly; no reference-state anchoring is required.

8 Summary of the Solve Structure

Step	**Type**	**Method**
P_{sat} (states 1, 3)	closed-form	Ambrose–Walton
Z (each state)	closed-form	cubic root (direct)
h, s (each state)	closed-form	Shomate + PR departure
State 2 discharge T	1-D root find	bracketed bisection (monotone)
State 4 quality	closed-form	linear interpolation
Full flowsheet (IDAES/IPOPT)	simultaneous solve	*not used here*

Every deviation case (Tc, Pc perturbed) is therefore evaluated deterministically: one answer per case, no convergence flag, no warmstart dependence, and no tolerance floor obscuring small perturbations.